

Note: The instruction to cover a range of 1 to 31 has to be observed strictly.

6(a)(i)

Potential difference across the lamp

$$V_{\text{lamp}} = IR = 0.30 \times 5.0 \\ = 1.5 \text{ V}$$

P.d. across R , $V_R = 1.5 \text{ V}$

Current through the circuit,

$$I = \frac{E - V_R}{3.0} = \frac{6.0 - 1.5}{3.0} \\ = 1.5 \text{ A}$$

$$R = \frac{V_R}{I_R} = \frac{1.5}{1.5 - 0.30} \\ = 1.25 \Omega$$

Note: A clear step-by-step approach in showing clear working is important.

No.	Solution
6(a)(ii)	Total power transferred to the circuit $P_{\text{total}} = VI$ $= 6.0 \times 1.5$ $= 9.0 \text{ W}$ Power transferred to the lamp $P_{\text{lamp}} = I^2 R$ $= (0.30)^2 \times 5.0$ $= 0.45 \text{ W}$ Energy transferred to the lamp $E_{\text{lamp}} = \frac{P_{\text{lamp}}}{P_{\text{total}}} \times E_{\text{total}}$ $= \frac{0.45}{9.0} \times 120$ $= 6.0 \text{ J}$
6(a)(iii)	The resistance of the filament in the lamp will be at its lowest value when the current is first switched on as the filament wire will be cool. Cool metal filament has lower resistances than when it is hot. Hence, the initial current flow in the filament will be greatest. As the current begins to flow, there will be increase in the filament temperature, causing increase in its resistance and decrease in current flow through it.
6(b)	Resistance wires X and Y are connected in series. Hence, same current flows through X and Y. Using $I = Anvq$, $n = \frac{I}{Avq}$ $= \frac{I}{\pi r^2 v q}$ $\therefore n \propto \frac{1}{r^2 v}$ ratio $\frac{\text{number density of charge carriers in Y}}{\text{number density of charge carriers in X}}$ $\frac{n_Y}{n_X} = \frac{v_X r_X^2}{v_Y r_Y^2}$ $= 3 \times \left(\frac{1}{2}\right)^2$ $= 0.75$

No.	Solution
7(a)	charge distribution